

TENSORS AND EXTERIOR FORMS

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ABSTRACT. This study note establishes the notion of a tensor, including both notions of vector and a whole class of objects characterized by a transformation law. Such definitions are learned "intrinsically." That is, in a basis-free fashion. *Notation is designed to help us recognize intrinsic quantities when they are presented in component form and to help prevent us from making blatant errors*

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1. COVECTORS AND RIEMANNIAN METRICS

To briefly state, this aims to address finding the *steepest ascent of something*. This may remind you of a slope, or something you came across of whilst learning EM.

1.1. Linear Functionals and the Dual Space

Again, we let $\mathbf{E}^3$ be a vector space, but let's just call it $\mathbf{E} = E$. We do this because we are no longer going to require $\mathbf{E}$ to be restricted to 3 dimensions. E could be infinite-dimensional, but for sanity, we concern ourselves with a finite-dimensional space. Note that the n -dimensional vector space E has no prescribed basis.

So, for E , we need to choose a basis. For an n -dimensional vector space E , choosing a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ allows any vector $\mathbf{v} \in E$ to be written uniquely as a linear combination:

$$\mathbf{v} = \sum_j \mathbf{e}_j v^j = \sum_j v^j \mathbf{e}_j$$

- **Components:** The n real numbers v^j are the components of $\mathbf{v}$ with respect to the chosen basis.
- **Notation:** Superscripts (v^j) are used for vector components rather than subscripts.

There is a deliberate choice in how these terms are ordered based on the context:

- **Algebraic Preference ($\mathbf{e}_j v^j$):** Placing the scalar components v^j to the **right** of the basis elements $\mathbf{e}_j$ is preferred for matrix algebra.
- **Matrix View:** This ordering aligns cleanly with standard matrix multiplication conventions when treating the basis as a row vector.
- **Calculus Caveat:** This notation can feel awkward when transitioning into calculus on $\mathbb{R}^n$, where standard functions usually place coefficients to the left.

Example 1.1. Using our favored presentation, a vector would be written as:

$$\mathbf{v} = \sum_j \frac{\partial}{\partial x^j} v^j$$

This makes it appear, incorrectly, that we are differentiating the components v^j . To prevent this confusion, we employ a bold ∂ to remind us that we are not differentiating the components in this expression. Alternatively, we will sometimes simply use the traditional $\sum_j v^j \mathbf{e}_j$.

We define the following matrices to formalize the algebraic presentation:

- **Basis Row Matrix:** $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ is a symbolic row matrix where each entry is a vector rather than a scalar.
- **Component Column Matrix:** $v = (v^1, \dots, v^n)^T$ preserves the traditional notation of representing vector components as a column matrix.

Using these definitions, we can write our preferred representation compactly as a matrix product:

$$\mathbf{v} = \mathbf{e}v \tag{1}$$

where $\mathbf{v}$ is treated formally as a 1×1 matrix.

An n -dimensional vector space E with a choice of basis is isomorphic to $\mathbb{R}^n$ under the correspondence:

$$\mathbf{v} \rightarrow (v^1, \dots, v^n) \in \mathbb{R}^n$$

However, this isomorphism is "**unnatural**" because it is entirely dependent on the specific choice of basis.

Definition 1.2. A (real) **linear functional** α on E is a real-valued linear function α , that is, a linear transformation $\alpha : E \rightarrow \mathbb{R}$ from E to the 1-dimensional vector space $\mathbb{R}$. Thus

$$\alpha(a\mathbf{v} + b\mathbf{w}) = a\alpha(\mathbf{v}) + b\alpha(\mathbf{w})$$

for real numbers a, b , and vectors $\mathbf{v}, \mathbf{w}$.

By induction, we have, for any basis $\mathbf{e}$

$$\alpha\left(\sum_j \mathbf{e}_j v^j\right) = \sum_j \alpha(\mathbf{e}_j) v^j \tag{2}$$

This is simply of the form $\sum a_j v^j$ (where $a_j := \alpha(\mathbf{e}_j)$), and this is a linear function of the components of $\mathbf{v}$. Clearly if $\{a_j\}$ are any real numbers, then $\mathbf{v} \mapsto \sum a_j v^j$ defines a linear functional on all of E . Thus, after one has picked a basis, the most general linear functional on the finite-dimensional vector space E is of the form

$$\alpha(\mathbf{v}) = \sum_j a_j v^j \quad \text{where } a_j := \alpha(\mathbf{e}_j) \tag{3}$$

Warning

A linear functional α on E is not itself a member of E ; that is, α is not to be thought of as a vector in E . This is especially obvious in infinite-dimensional cases.

Example 1.3. Let E be the vector space of all continuous real-valued functions $f : \mathbb{R} \rightarrow \mathbb{R}$ of a real variable t . The **Dirac functional** δ_0 is the linear functional on E defined by

$$\delta_0(f) = f(0) \quad (4)$$

You should convince yourself that E is a vector space and that δ_0 is a linear functional.

Definition 1.4. The collection of all linear functionals α on a vector space E form a new vector space E^* , the **dual space** to E , under the operations

$$\begin{aligned} (\alpha + \beta)(\mathbf{v}) &:= \alpha(\mathbf{v}) + \beta(\mathbf{v}), \quad \alpha, \beta \in E^*, \quad \mathbf{v} \in E \\ (c\alpha)(\mathbf{v}) &:= c\alpha(\mathbf{v}), \quad c \in \mathbb{R} \end{aligned}$$

If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of E , we define the **dual basis** $\{\sigma^1, \dots, \sigma^n\}$ of E^* by setting their values on the basis vectors using the Kronecker delta:

$$\sigma^i(\mathbf{e}_j) = \delta_j^i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (5)$$

Extending σ^i by linearity to any vector $\mathbf{v} = \sum_j \mathbf{e}_j v^j \in E$ yields:

$$\sigma^i \left(\sum_j \mathbf{e}_j v^j \right) = \sum_j \sigma^i(\mathbf{e}_j) v^j = \sum_j \delta_j^i v^j = v^i \quad (6)$$

Hence, σ^i is the linear functional that extracts the i -th component of a vector $\mathbf{v}$ with respect to the basis $\mathbf{e}$.

- **Linear Independence:** Assume a linear combination equals the zero functional: $\sum_j a_j \sigma^j = 0$. Applying this to a basis vector $\mathbf{e}_k$:

$$0 = \sum_j a_j \sigma^j(\mathbf{e}_k) = \sum_j a_j \delta_k^j = a_k$$

Since $a_k = 0$ for all k , the functionals are linearly independent.

- **Spanning Set:** Let $\alpha \in E^*$. Evaluating α on an arbitrary vector $\mathbf{v}$ yields:

$$\alpha(\mathbf{v}) = \alpha \left(\sum_j \mathbf{e}_j v^j \right) = \sum_j \alpha(\mathbf{e}_j) v^j = \sum_j \alpha(\mathbf{e}_j) \sigma^j(\mathbf{v}) = \left(\sum_j \alpha(\mathbf{e}_j) \sigma^j \right) (\mathbf{v})$$

Since this holds for all $\mathbf{v}$, any linear functional can be decomposed uniquely as:

$$\alpha = \sum_j \alpha(\mathbf{e}_j) \sigma^j \quad (7)$$

Let $a_j = \alpha(\mathbf{e}_j)$ define the j -th component of the functional α . We can represent these components and the dual basis elements via matrices:

- **Dual Basis Column Matrix:** $\sigma = (\sigma^1, \dots, \sigma^n)^T$
- **Component Row Matrix:** $a = (a_1, \dots, a_n)$

This allows us to express the functional compactly as a matrix product:

$$\alpha = \sum_j a_j \sigma^j = a\sigma \quad (8)$$

While vector components are written as column matrices (v), the components of a linear functional are written as row matrices (a).

Let $\beta = (\beta_{iR})$ be a matrix whose entries are linear functionals, and let $\mathbf{f} = (\mathbf{f}_{Rs})$ be a matrix whose entries are vectors. The matrix product $\beta\mathbf{f}$ (or $\beta(\mathbf{f})$) produces a matrix of scalars defined by:

$$\beta(\mathbf{f})_{is} := \sum_R \beta_{iR}(\mathbf{f}_{Rs}) \quad (9)$$

Recall that the dual basis matrix $\sigma = (\sigma^1, \dots, \sigma^n)^T$ is a column matrix of functionals, and the basis matrix $\mathbf{e} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ is a row matrix of vectors.

Since $\sigma^i(\mathbf{e}_j) = \delta_j^i$, their formal matrix product $\sigma\mathbf{e}$ yields the $n \times n$ identity matrix:

$$\sigma\mathbf{e} = I_n \quad (10)$$

Using this property, we can express the action of a functional α on a vector $\mathbf{v}$ entirely via associative matrix multiplication:

$$\alpha(\mathbf{v}) = (a\sigma)(\mathbf{e}v) = a(\sigma\mathbf{e})v = aI_nv = av \quad (11)$$

where a is the component row matrix of the functional, and v is the component column matrix of the vector.

1.2. The Differential of a Function

Definition 1.5. The dual space M_p^{n*} to the tangent space M_p^n at the point p of a manifold is called the **cotangent space**.

Definition 1.6. Let $f : M^n \rightarrow \mathbb{R}$. The **differential** of f at p , written df , is the linear functional $df : M_p^n \rightarrow \mathbb{R}$ defined by

$$df(\mathbf{v}) = \mathbf{v}_p(f) \quad (12)$$

The differential df is defined intrinsically, independent of any choice of coordinates or basis. However, introducing local coordinates allows us to express it explicitly in components.

In local coordinates, the standard coordinate basis vectors for the tangent space M_p^n are given by $\mathbf{e}_j = \partial/\partial x^j|_p$. Evaluating the differential df on a tangent vector $\mathbf{v} = \sum v^j \frac{\partial}{\partial x^j}$ yields:

$$df \left(\sum_j v^j \frac{\partial}{\partial x^j} \right) = \sum_j v^j(p) \frac{\partial f}{\partial x^j}(p) \quad (13)$$

This confirms that df acts linearly on the component functions v^j .

If we choose the function f to be a coordinate function x^i , its differential dx^i acts on a coordinate basis vector as follows:

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \frac{\partial x^i}{\partial x^j} = \delta_j^i \quad (14)$$

By linearity, when dx^i acts on a general tangent vector, it extracts exactly its i -th component:

$$dx^i \left(\sum_j v^j \frac{\partial}{\partial x^j} \right) = \sum_j v^j dx^i \left(\frac{\partial}{\partial x^j} \right) = v^i \quad (15)$$

Because $\sigma^i = dx^i$ satisfies the defining relation $\sigma^i(\mathbf{e}_j) = \delta_j^i$, the set of coordinate differentials $\{dx^1, \dots, dx^n\}$ forms the **dual basis** to the coordinate tangent basis $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$.

- Consequently, $\{dx^1, \dots, dx^n\}$ forms a basis for the cotangent space M_p^{n*} .

Using this natural basis, any general linear functional (covector) $\alpha \in M_p^{n*}$ can be expanded in components as:

$$\alpha = \sum_j \alpha \left(\frac{\partial}{\partial x^j} \right) dx^j = \sum_j a_j dx^j \quad (16)$$

where $a_j := \alpha \left(\frac{\partial}{\partial x^j} \right)$ are the components of the covector.

Eq. (16) is known as a **differential form**.

The modern definition of a differential updates elementary calculus to work on curved spaces. In standard $\mathbb{R}^3$ with Cartesian coordinates, the differential dx of an independent variable is classically treated as a simple coordinate difference between two points, P and Q :

$$dx = x' - x \quad (17)$$

This classical approach implicitly defines dx using a displacement vector $\mathbf{v} = Q - P$:

$$dx(\mathbf{v}) = dx(Q - P) = x' - x \quad (18)$$

While this calculation works in flat spaces like $\mathbb{R}^3$ where vectors can be defined by subtracting coordinates of endpoints, it fails on general manifolds. Because you cannot subtract points on a curved surface, modern differential geometry shifts perspectives: dx is treated strictly as a linear functional acting on a vector space attached to a specific point, rather than a gap between coordinates.

In elementary calculus, the total differential df of a dependent variable $f(x, y, z)$ is defined by combining its partial derivatives with the independent differentials:

$$\left(\frac{\partial f}{\partial x} \right) dx + \left(\frac{\partial f}{\partial y} \right) dy + \left(\frac{\partial f}{\partial z} \right) dz \quad (19)$$

The modern manifold definition unifies this concept. It treats independent and dependent variables identically, applying universally across any manifold. Evaluating the coordinate expression for df reveals that it matches the classical formula exactly when expanded in local coordinates:

$$df = \sum_j df \left(\frac{\partial}{\partial x^j} \right) dx^j = \sum_j \left(\frac{\partial f}{\partial x^j} \right) dx^j \quad (20)$$

A linear functional living on the tangent space ($\alpha : M_p^n \rightarrow \mathbb{R}$) is referred to as a **covariant vector**, **covector**, or **1-form**. When we smoothly assign a covector to every point across an open region of a manifold, it creates a **covector field**. This field can be expressed locally by letting its components vary as functions of position:

$$\alpha = \sum_j a_j(x) dx^j \quad (21)$$

The total differential df serves as a primary example of such a covector field.

When switching between two sets of overlapping local coordinate systems, say U and V , the basis 1-forms transform via the multivariate chain rule:

$$dx_V^i = \sum_j \left(\frac{\partial x_V^i}{\partial x_U^j} \right) dx_U^j \quad (22)$$

Because a general covector α is an intrinsic geometric object independent of coordinates, its expansions in both coordinate systems must be identical ($\sum_i a_i^V dx_V^i = \sum_j a_j^U dx_U^j$). Substituting the basis transformation into this invariance condition yields the transformation behavior for the components:

$$a_j^U = \sum_i a_i^V \left(\frac{\partial x_V^i}{\partial x_U^j} \right) \quad (23)$$

By utilizing the fact that the Jacobian matrices $\partial x_U / \partial x_V$ and $\partial x_V / \partial x_U$ are inverses of each other, we can invert this relationship. This yields the standard transformation law for the components of a covariant vector (or covector):

$$a_i^V = \sum_j a_j^U \left(\frac{\partial x_U^j}{\partial x_V^i} \right) \quad (24)$$

In shorthand matrix notation, if we denote the coordinate transition matrix as $c_{UV} = \frac{\partial x_U}{\partial x_V}$, the components transform directly with the transition matrix:

$$a^V = a^U c_{UV} = a^U c_{VU}^{-1} \quad (25)$$

This behavior defines a **covariant** transformation rule, as the components transform in the same manner as the coordinate basis elements themselves.

1.3. Scalar Products in Linear Algebra

An inner product (or scalar product) $\langle \mathbf{v}, \mathbf{w} \rangle$ on an n -dimensional vector space E is a symmetric, bilinear mapping that outputs a real number. It is non-degenerate, meaning that if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all vectors $\mathbf{w}$, then $\mathbf{v}$ must be the zero vector. While standard inner products are positive-definite ($\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all non-zero vectors), this requirement is relaxed here to accommodate applications in relativity, where spacetime metrics can have negative or zero intervals for non-zero vectors.

Expanding two vectors in a given basis as $\mathbf{v} = \sum_i v^i \mathbf{e}_i$ and $\mathbf{w} = \sum_j w^j \mathbf{e}_j$ allows us to extract components from the bilinear inner product:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i,j} v^i \langle \mathbf{e}_i, \mathbf{e}_j \rangle w^j \quad (26)$$

To represent this structure algebraically, we introduce the symmetric matrix $G = (g_{ij})$, whose components are determined by evaluating the inner product of the basis elements themselves:

$$g_{ij} := \langle \mathbf{e}_i, \mathbf{e}_j \rangle \quad (27)$$

This matrix (g_{ij}) is known as the metric tensor. It allows the inner product of any two vectors to be evaluated explicitly via their components:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i,j} v^i g_{ij} w^j \quad (28)$$

In standard matrix algebra, this is written as a matrix multiplication involving the component column matrices v and w :

$$\langle \mathbf{v}, \mathbf{w} \rangle = v^T G w \quad (29)$$

In the specific case where the chosen basis is orthonormal, the metric tensor simplifies to the identity matrix ($g_{ij} = \delta_{ij}$). This collapses the metric expression down to the standard Euclidean dot product formula $\sum_i v^i w^i$ typically used in elementary linear algebra.

Because the inner product is linear in its second slot when the first vector $\mathbf{v}$ is held fixed, every fixed vector $\mathbf{v} \in E$ naturally induces a linear functional ν acting on E :

$$\nu(\mathbf{w}) = \langle \mathbf{v}, \mathbf{w} \rangle \quad (30)$$

This establishes a fundamental connection where the metric tensor maps vectors directly into covectors.

The mapping induced by the non-degenerate inner product associates each vector $\mathbf{v} \in E$ with a unique linear functional $\nu \in E^*$. This dual functional is called the **covariant version** of the vector $\mathbf{v}$. We can expand this functional in terms of the dual basis σ by evaluating its components:

$$\nu = \sum_j \nu(\mathbf{e}_j) \sigma^j = \sum_j \langle \mathbf{v}, \mathbf{e}_j \rangle \sigma^j \quad (31)$$

By substituting the vector expansion $\mathbf{v} = \sum_i \mathbf{e}_i v^i$ into the inner product, the linearity of the metric allows us to compute the dual components directly:

$$\nu = \sum_j \left(\sum_i \langle \mathbf{e}_i v^i, \mathbf{e}_j \rangle \right) \sigma^j = \sum_j \left(\sum_i v^i g_{ij} \right) \sigma^j \quad (32)$$

In standard tensor notation, it is customary to drop the Greek letter ν and denote the components of this covariant version using the same letter v , distinguished by a lower index. This calculation relates the lower covector components directly to the upper vector components:

$$v_j = \sum_i v^i g_{ij} = \sum_i g_{ji} v^i \quad (33)$$

This index configuration forms the foundation of index gymnastics. This algebraic step is called "**lowering the index**" using the metric tensor g_{ij} , transforming the contravariant components v^i into the covariant components v_j .

If the chosen basis happens to be completely orthonormal, the metric components reduce to the Kronecker delta ($g_{ij} = \delta_{ij}$). Under these specific geometric conditions, the index-lowering operation has no numerical effect, meaning the covariant components match the contravariant components identically ($v_j = v^j$).

In a finite-dimensional inner product space E , the relationship between vectors and covectors is entirely bijective due to the non-degeneracy of the metric. Every linear functional $\nu = \sum_j v_j \sigma^j$ can be mapped uniquely back to an underlying vector $\mathbf{v}$ satisfying $\nu(\mathbf{w}) = \langle \mathbf{v}, \mathbf{w} \rangle$. Since the metric matrix $G = (g_{ij})$ is non-degenerate, its matrix inverse G^{-1} is guaranteed to exist and remain symmetric. We denote the entries of this inverse matrix using upper indices:

$$G^{-1} = (g^{ij}) \quad (34)$$

By inverting the index-lowering relations, we can compute the contravariant components v^i of the vector directly from the given covariant components v_j :

$$v^i = \sum_j g^{ij} v_j \quad (35)$$

This operation is called **"raising the index"** using the inverse metric tensor g^{ij} , yielding the contravariant components of the covector.

Evaluating a vector's components explicitly highlights the geometric role of the metric. The covariant components are given directly by projecting the vector onto the basis elements:

$$v_j = \langle \mathbf{v}, \mathbf{e}_j \rangle \quad (36)$$

Substituting this inner product into the index-raising formula yields a general expansion for any vector $\mathbf{v}$ in an arbitrary, non-orthogonal basis:

$$\mathbf{v} = \sum_i \left(\sum_j g^{ij} \langle \mathbf{v}, \mathbf{e}_j \rangle \right) \mathbf{e}_i \quad (37)$$

This identity generalizes the traditional Euclidean expansion $\mathbf{v} = \sum_i \langle \mathbf{v}, \mathbf{e}_i \rangle \mathbf{e}_i$, which is a special case that holds only when the chosen basis is completely orthonormal. In a non-orthogonal coordinate system, such as a skewed coordinate system on the plane $\mathbb{R}^2$ spanning unit vectors that are not perpendicular, the inverse metric terms g^{ij} act as essential correction factors to properly account for the geometric tilt of the basis.

For any n -dimensional vector space E , the dual vector space E^* can always be constructed intrinsically without referencing a basis or requiring an inner product. While choosing a basis $\mathbf{e}$ for E automatically determines a dual basis σ for E^* , the resulting component-wise bijection ($\sum a_j \sigma^j \rightarrow \sum a_j \mathbf{e}_j$) is highly dependent on that specific coordinate choice. Because this 1:1 mapping changes whenever the underlying basis is altered, it is considered an **"unnatural isomorphism"** between E and E^* .

Introducing an inner product onto E provides an entirely different, basis-independent correspondence between the two spaces. This natural mapping pairs a covector $\nu \in E^*$ with the unique vector $\mathbf{v} \in E$ that satisfies the inner product relation:

$$\nu = \langle \mathbf{v}, \cdot \rangle \quad (38)$$

Under this geometric framework, each individual dual basis element σ^i can be uniquely represented by an associated vector $\mathbf{f}_i$ such that:

$$\sigma^i(\mathbf{w}) = \langle \mathbf{f}_i, \mathbf{w} \rangle \quad (39)$$

The collection of these vectors $\mathbf{f} = \{\mathbf{f}_i\}$ forms a new basis for the original vector space E . This set is referred to as the **basis of E dual to $\mathbf{e}$** , and by definition satisfies the orthogonality relation:

$$\langle \mathbf{f}_i, \mathbf{e}_j \rangle = \delta_j^i \quad (40)$$

Although this dual vector basis approach is frequently used in branches of applied mathematics (such as solid-state physics and engineering), differential geometry avoids identifying covectors with vectors in this manner. Preserving the distinction between the two spaces is crucial because covectors possess a specialized, powerful calculus—such as exterior differentiation and integration over surfaces—that cannot be naturally applied to standard vectors.

1.4. Riemannian Manifolds and the Gradient Vector

A **Riemannian metric** on a manifold M^n assigns a smoothly varying, positive-definite inner product $\langle \cdot, \cdot \rangle$ to each tangent space M_p^n . If the inner product is non-degenerate but not positive-definite, it is called a **pseudo-Riemannian metric**, which defines a (pseudo-) Riemannian manifold.

In a coordinate basis where $\mathbf{e}_i = \partial_i = \partial/\partial x^i$, the components of the metric tensor vary as functions of position:

$$g_{ij}(x) = \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle \quad (41)$$

When switching coordinate systems on an overlap region $U \cap V$, applying the multivariate chain rule to the coordinate basis vectors reveals how the metric components transform:

$$g_{ij}^V = \sum_{r,s} \left(\frac{\partial x_U^r}{\partial x_V^i} \right) \left(\frac{\partial x_U^s}{\partial x_V^j} \right) g_{rs}^U \quad (42)$$

This establishes the standard tensor transformation law for a covariant tensor of rank 2.

Definition 1.7. Let M^n be a (pseudo-) Riemannian manifold and f be a differentiable function. The **gradient vector**, denoted $\text{grad } f$ or ∇f , is the contravariant vector field associated with the covector df via the metric:

$$df(\mathbf{w}) = \langle \nabla f, \mathbf{w} \rangle \quad (43)$$

By converting the coordinate components using the inverse metric tensor to raise the index, we obtain the explicit coordinate expression for the components of the gradient vector:

$$(\nabla f)^i = \sum_j g^{ij} \frac{\partial f}{\partial x^j} \quad (44)$$

The squared norm of the gradient vector can be computed by evaluating the inner product of the gradient with itself, which is equivalent to letting the differential df act directly on the gradient vector:

$$\|\nabla f\|^2 = \langle \nabla f, \nabla f \rangle = df(\nabla f) = \sum_{i,j} \frac{\partial f}{\partial x^i} g^{ij} \frac{\partial f}{\partial x^j} \quad (45)$$

If the coordinate system yields a standard Euclidean metric ($g^{ij} = \delta_i^j$), the components of the gradient vector $(\nabla f)^i$ match the standard partial derivatives $\partial f/\partial x^i$ from elementary calculus.

1.5. Curves of Steepest Ascent

The gradient vector on a Riemannian manifold shares the same geometric interpretation as it does in Euclidean space. If $\mathbf{v}$ is a unit tangent vector at a point $p \in M$, the directional derivative of f in the direction of $\mathbf{v}$ is given by:

$$\mathbf{v}(f) = \sum_j \left(\frac{\partial f}{\partial x^j} \right) v^j = df(\mathbf{v}) = \langle \nabla f, \mathbf{v} \rangle \quad (46)$$

For a positive-definite inner product, applying the Cauchy-Schwarz inequality yields $|\mathbf{v}(f)| = |\langle \nabla f, \mathbf{v} \rangle| \leq \|\nabla f\| \|\mathbf{v}\| = \|\nabla f\|$. This inequality demonstrates that the function f achieves its maximum rate of change along the direction pointed to by the gradient vector ∇f .

The level set of f passing through a point p with value $f(p) = a$ is defined as the subset:

$$M^{n-1}(a) := \{x \in M^n \mid f(x) = a\} \quad (47)$$

If the differential df does not vanish at p , this level set forms an $(n-1)$ -dimensional sub manifold in a local neighborhood of p . For any curve $x = x(t)$ lying entirely within this level set and passing

through p , the function value $f(x(t)) = a$ remains constant. Differentiating both sides with respect to time shows that the velocity tangent vector dx/dt is annihilated by the differential:

$$df \left(\frac{dx}{dt} \right) = \frac{d}{dt} f(x(t)) = 0 \quad (48)$$

While it is tempting to state that the differential df is orthogonal to the level set, this statement is geometrically imprecise because df is a covector rather than a vector. Instead, its contravariant vector equivalent—the gradient ∇f —is strictly orthogonal to the tangent space of the level set:

$$\left\langle \nabla f, \frac{dx}{dt} \right\rangle = df \left(\frac{dx}{dt} \right) = 0 \quad (49)$$

Because a contravariant vector cannot be directly equated to a covariant differential ($dx^i/dt \neq \partial f/\partial x^i$), setting up a physically valid vector flow requires using the gradient. The local differential equations defining the integral curves tangent to the gradient vector are given by:

$$\frac{dx^i}{dt} = (\nabla f)^i = \sum_j g^{ij} \left(\frac{\partial f}{\partial x^j} \right) \quad (50)$$

These curves track the path of steepest ascent. Along these specific curves, the rate of change of the function value scales with the squared norm of the gradient:

$$\frac{df}{dt} = df \left(\frac{dx}{dt} \right) = \langle \nabla f, \nabla f \rangle = \|\nabla f\|^2 \quad (51)$$

If we instead parameterize the system to normalize the speed of ascent along these trajectories by modifying the differential equations to:

$$\frac{dx}{dt} = \frac{\nabla f}{\|\nabla f\|^2} \quad (52)$$

The rate of change simplifies to a constant value: $df/dt = 1$. Under this normalized flow, the time parameter t required to travel from an initial level set $f = a$ to a subsequent level set $f = a + t$ matches the shift in value exactly. This local transformation of a manifold's level sets is termed a Morse deformation.

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